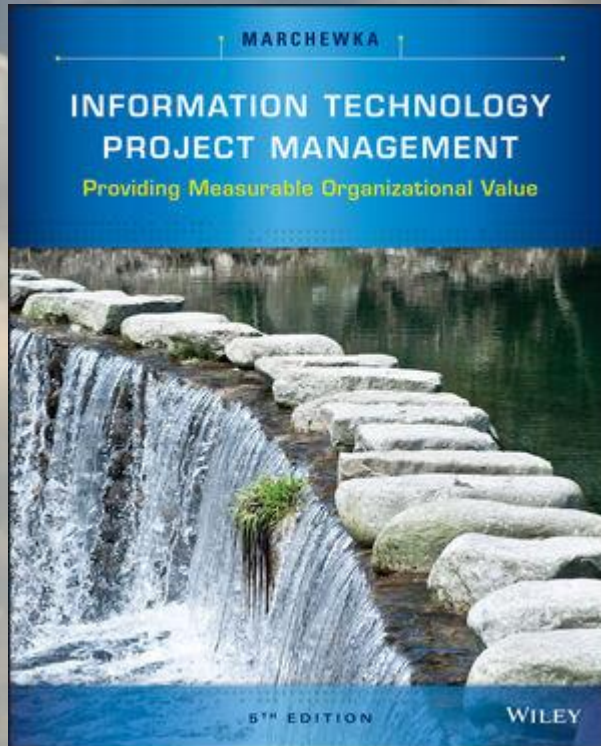


COMo463 IT Project Management

Spring 2025

Information Technology Project Management 4e

Chapter 8-Project Risk Management



Time	Lecture	Note
14:00-14:45	Hour-1	Theory & Practices
15:00-15:45	Hour-2	Theory & Practices
16:00-16:45	Hour-3	Theory & Practices
17:00-17:45	Hour-4	Theory & Practices



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Computer
Engineering



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Learning Objectives



Risk Management

PMBOK



Risk management plan

Describe how to plan risk management



Project risks

Identification

Causes

Effects



Risk Analysis

Qualitative techniques

Quantitative techniques



Prioritization

Analyse project risks

Prioritize



Risk strategies

insurance

avoidance

mitigation



Describe

risk monitoring

control



Risk evaluation

To learn from experience

To identify best practices



PMBOK

Risk Management

Knowledge Areas	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
[4] Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	•4.3 Direct and Manage Project Work •4.4 Manage Project Knowledge	•4.5 Monitor and Control Project Work •4.6 Perform Integrated Change Control	4.7 Close Project or Phase
[5] Project Scope Management		•5.1 Plan Scope Management •5.2 Collect Requirements •5.3 Define Scope •5.4 Create WBS		•5.5 Validate Scope •5.6 Control Scope	
[6] Project Schedule Management		•6.1 Plan Schedule Management •6.2 Define Activities •6.3 Sequence Activities •6.4 Estimate Activity Durations •6.5 Develop Schedule		6.6 Control Schedule	
[7] Project Cost Management		•7.1 Plan Cost Management •7.2 Estimate Costs •7.3 Determine Budget		7.4 Control Costs	
[8] Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	
[9] Project Resource Management		•9.1 Plan Resource Management •9.2 Estimate Activity Resources	•9.3 Acquired Resources •9.4 Develop Team •9.5 Manage Team	9.6 Control Resources	
[10] Project Communications Management		10.1 Plan Communications Management	10.2 Manage Communications	10.3 Monitor Communications	
[11] Project Risk Management		•11.1 Plan Risk Management •11.2 Identify Risks •11.3 Perform Qualitative Risk Analysis •11.4 Perform Quantitative Risk Analysis •11.5 Plan Risk Responses	11.6 Implement Risk Responses	11.7 Monitor Risks	
[12] Project Procurement Management		12.1 Plan Procurement Management	12.2 Conduct Procurements	12.3 Control Procurements	
[13] Project Stakeholder Management	13.1 Identify Stakeholders	13.2 Plan Stakeholder Engagement	13.3 Manage Stakeholder Engagement	13.4 Monitor Stakeholder Engagement	

PI

Ide

Risk
Managem
Planni



PMBOK® Risk Management Processes: Identification



Identify Risks

which risks can impact
the Project?



Include

project stakeholders



requires

understanding of the
project's objectives

Identification: PMBOK® Project Risks



Goal

Risks against
reaching the project
goal



Scope

Risks against
completing the project
scope



Schedule

Risks against
Catching the schedule



Budget

Risks cause
Budget overrun



Quality

Risks to
Reduce product quality



PMBOK® Risk Management Processes: Analysis



Risk analysis

Qualitative

Quantitative



Qualitative

concerning

the impact

the likelihood



Quantitative

developing a probabilistic
model

to understand and
respond

PMBOK® Risk Management Processes: Response



Response Planning

Developing procedures
and techniques



Responses

Reduce threats
Enhance opportunities



Reduce

the threats of risks



enhance

the likelihood of
opportunities

PMBOK® Risk Management Processes: Monitoring & control



Risk Monitoring and Control

Providing an early warning system



for

Identified risks
new risks



ensure risk responses

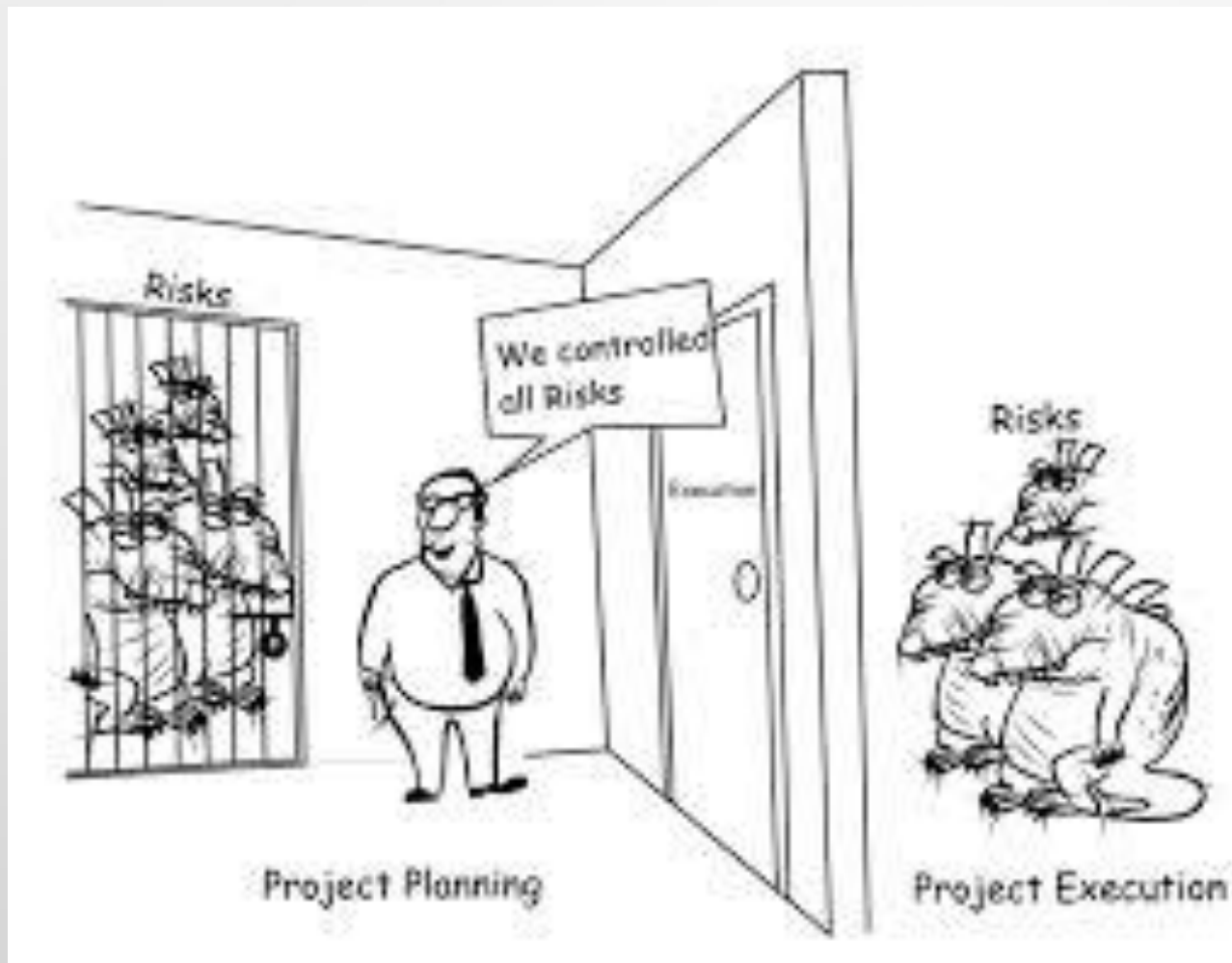
to be implemented as planned
had the effect as intended.



Risk Management

Common Mistakes

New risks can arrive anytime

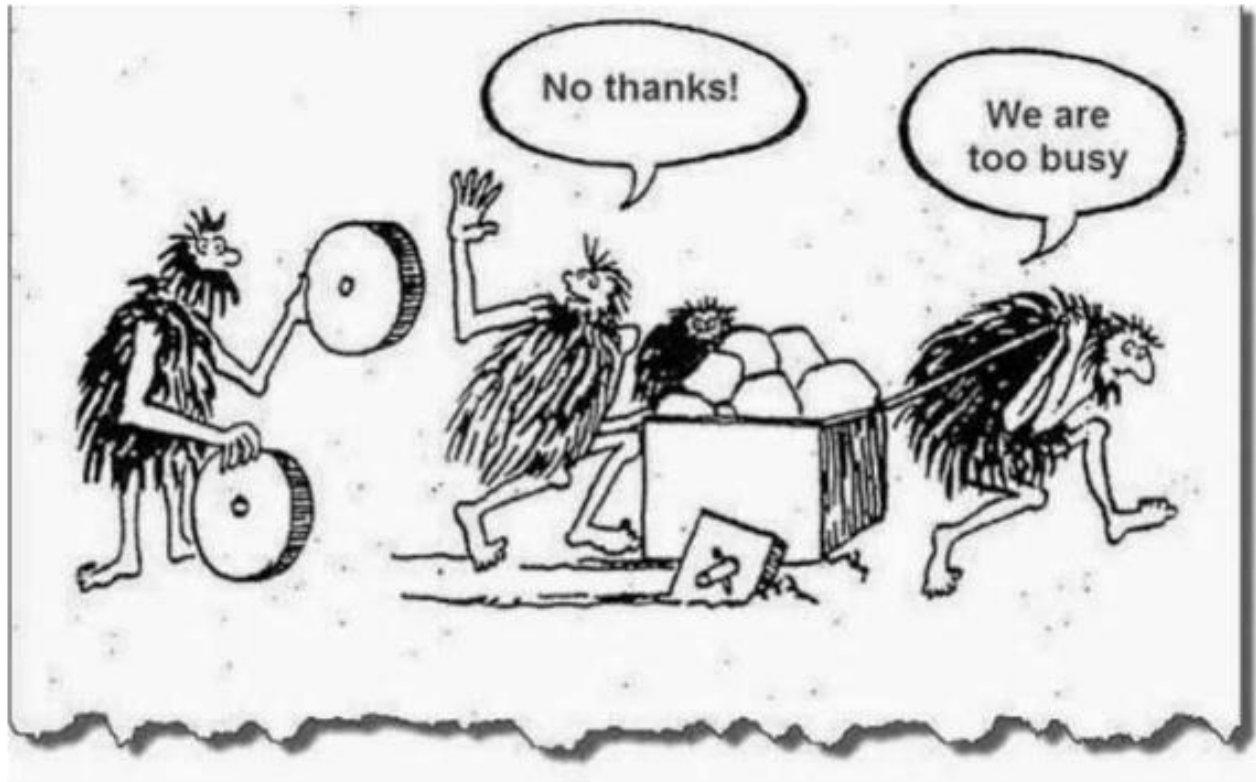


Risk Management Has Benefits



Reserve Adequate time for risk management

... so we end up behind schedule



Geir Amsjøl

5

Identify and assess risks using a standardized approach



Effective Risk Management Conditions



Commitment by all
stakeholders



Stakeholder
responsibility



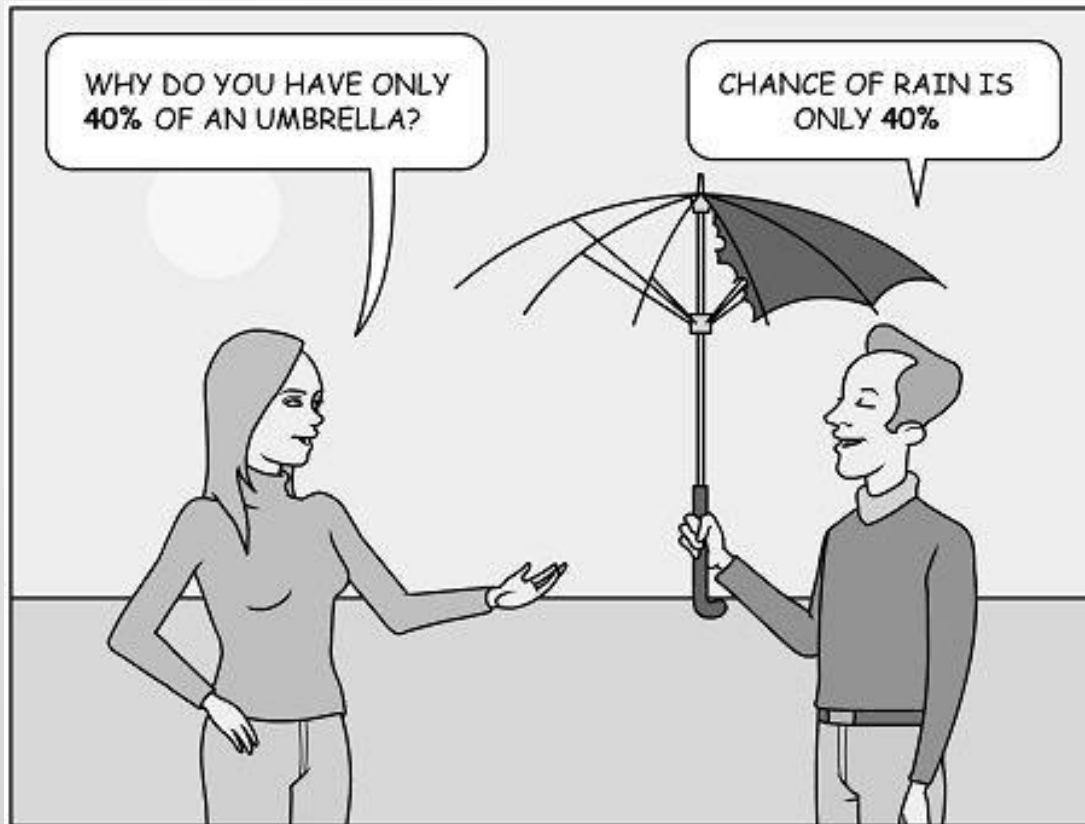
Different risks for
different types of projects

Definitions

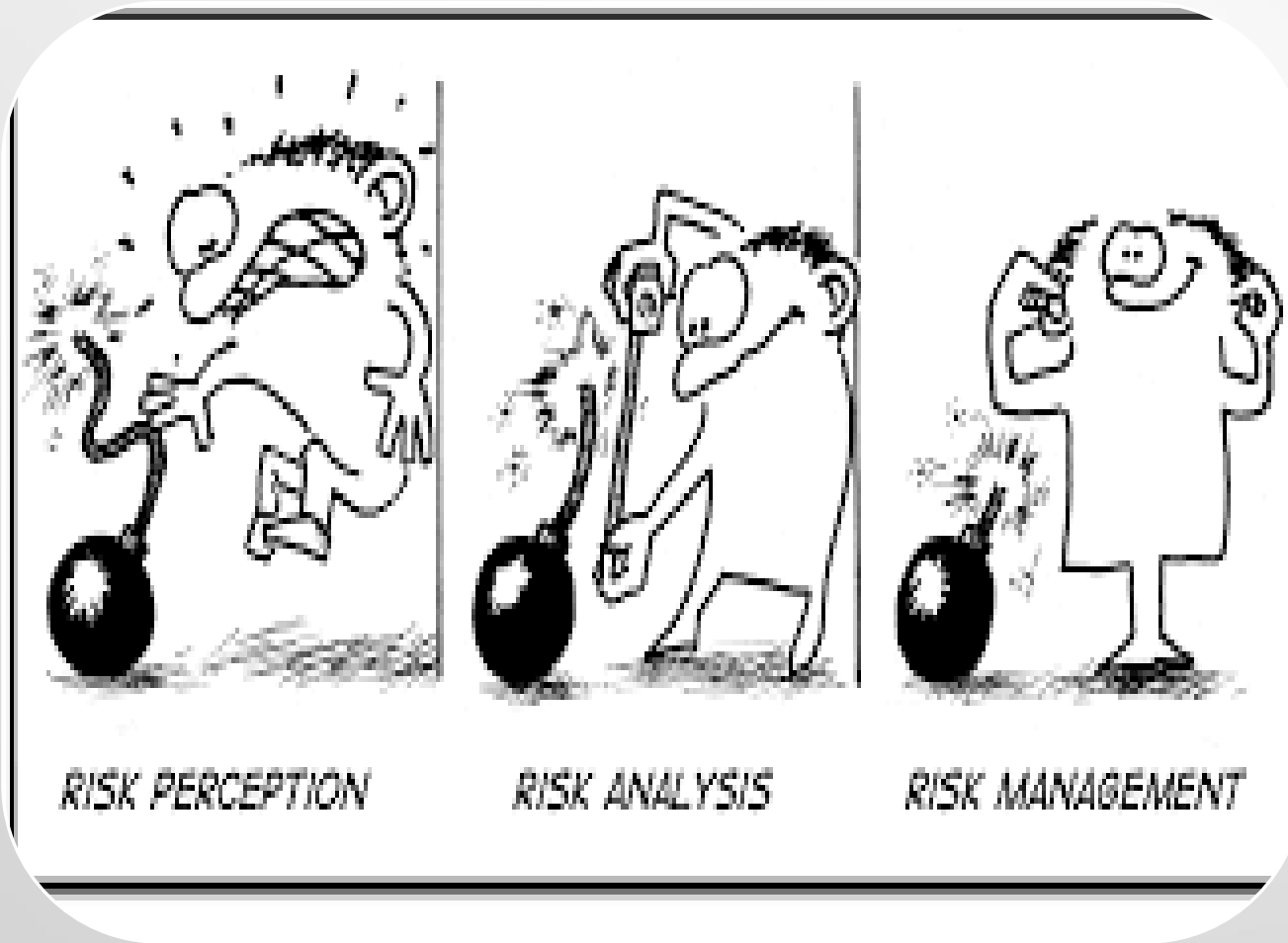
Risk

An uncertain event or condition

has a positive or negative effect on the project objectives.



Risk: TERMINOLOGY

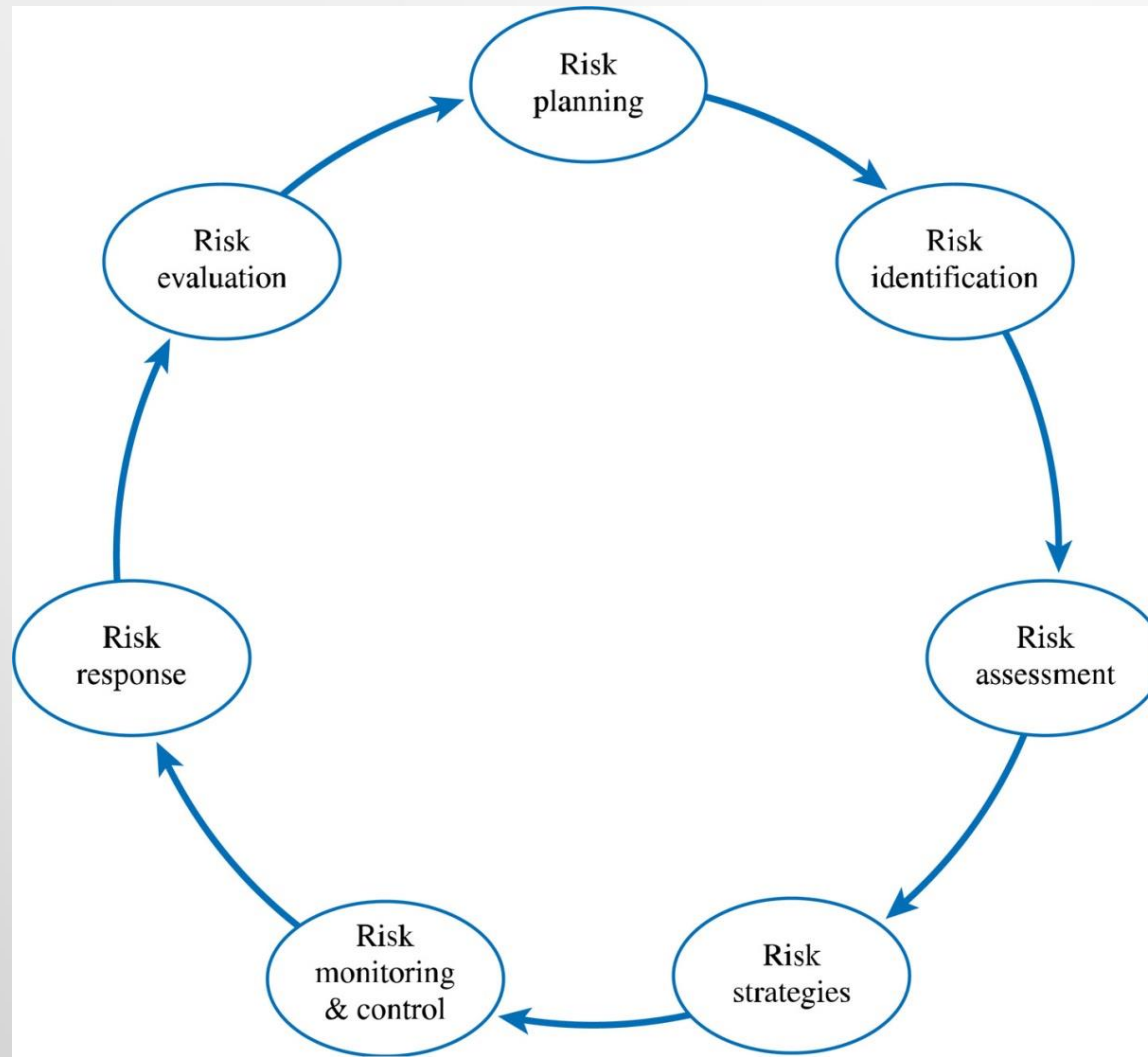




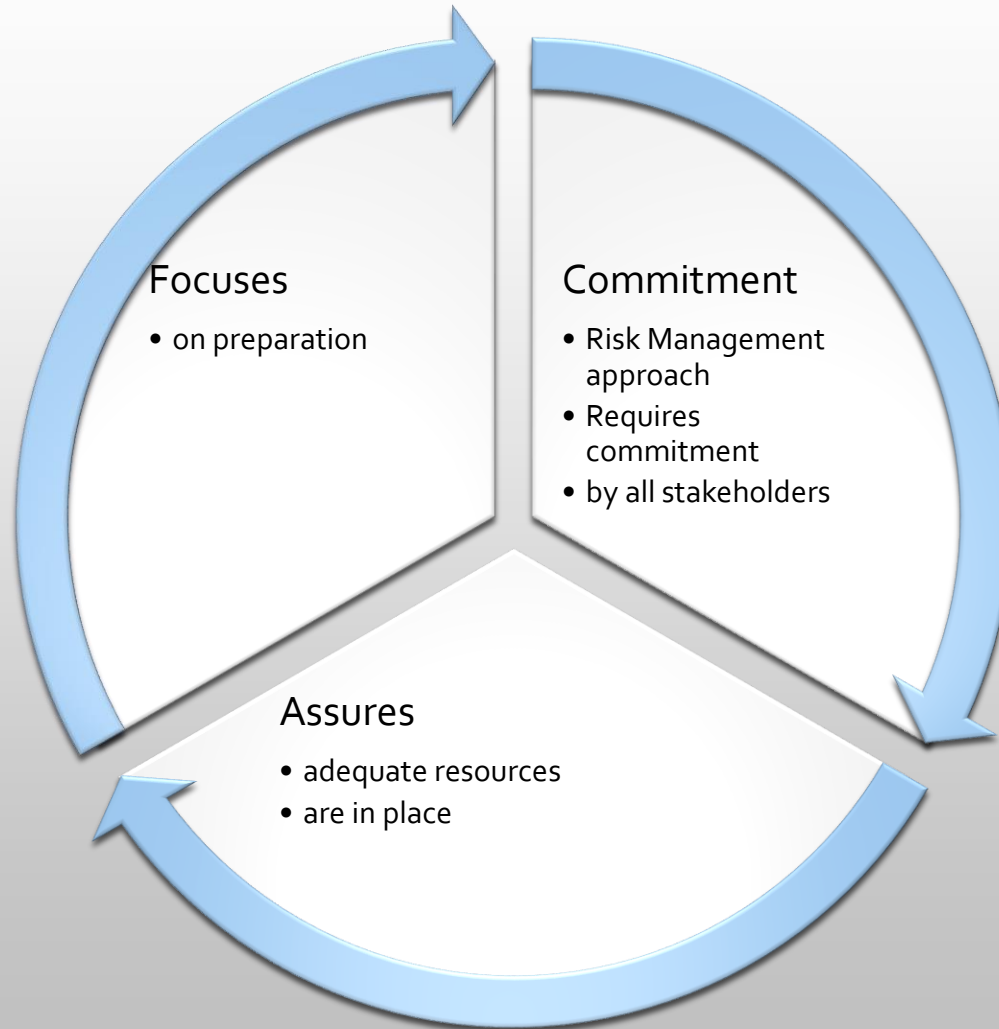
Processes

IT Project Risk Management

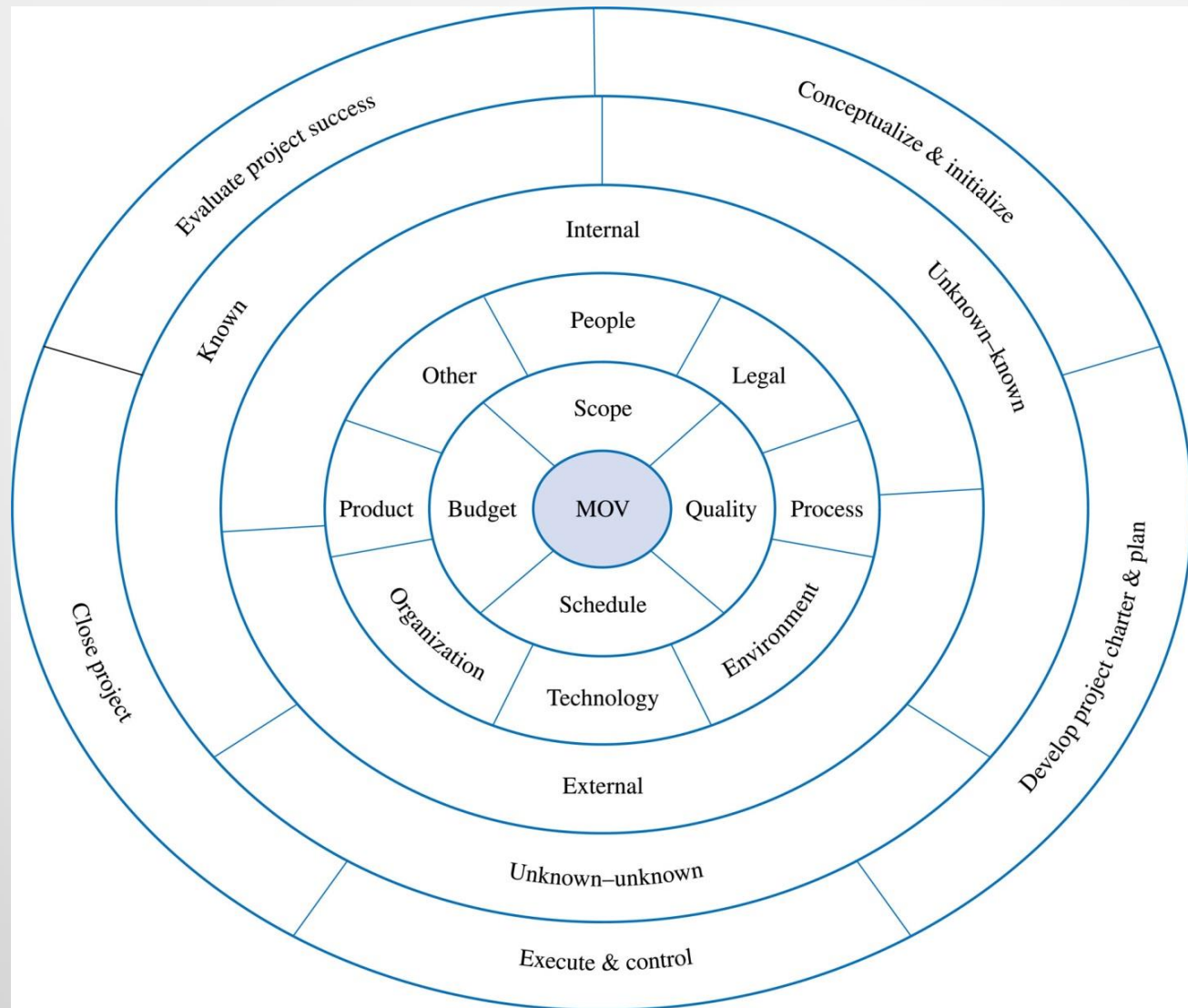
IT Project Risk Management Processes



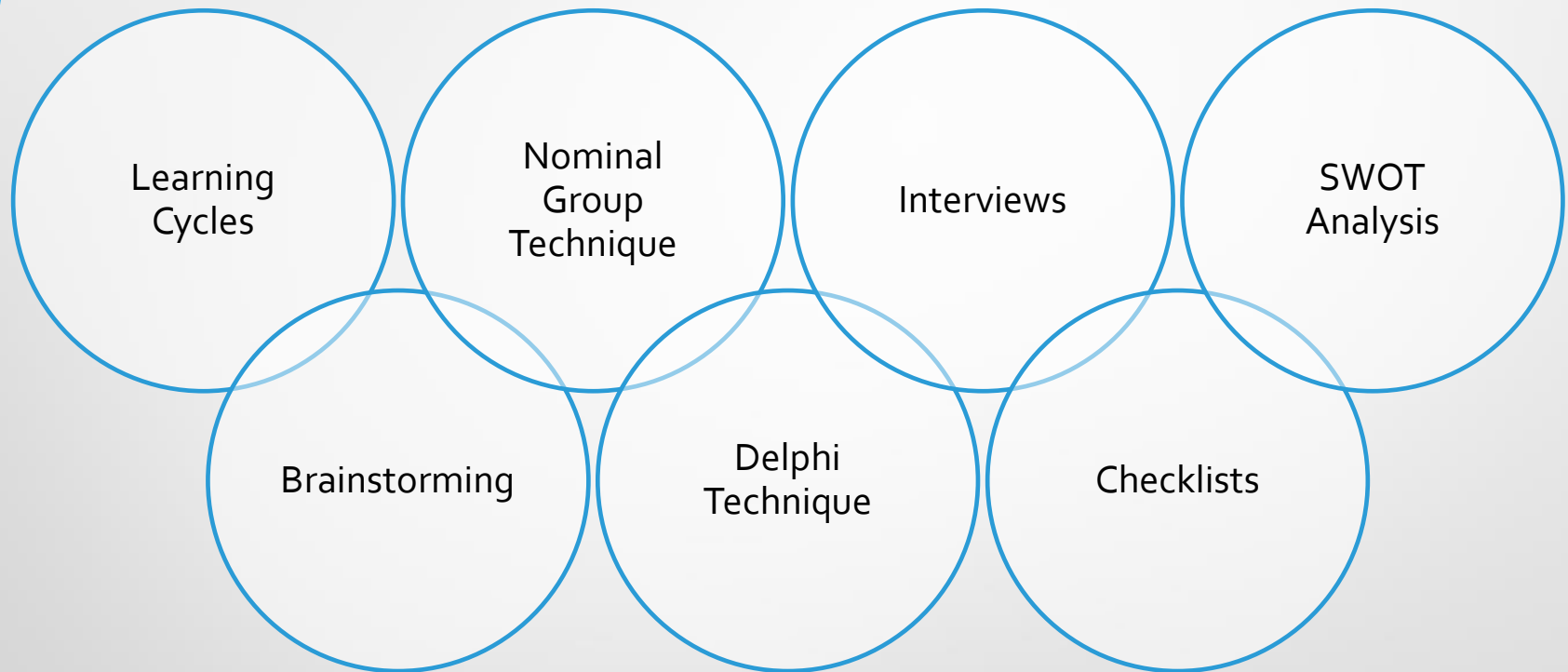
Risk Planning



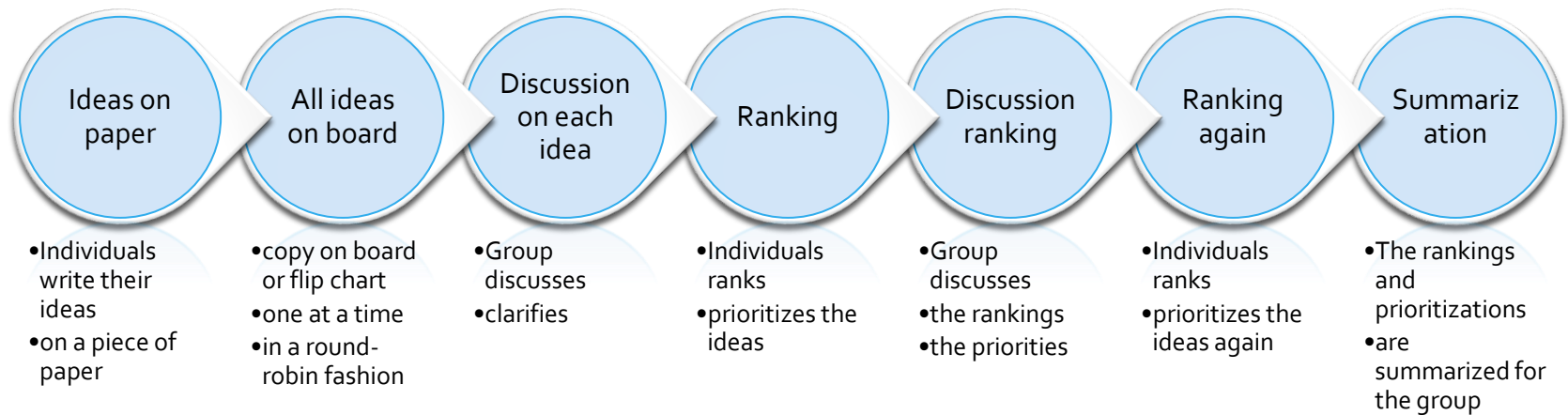
IT Project Risk Identification Framework



Risk Identification Tools & Techniques



NOMINAL GROUP TECHNIQUE (NGT)

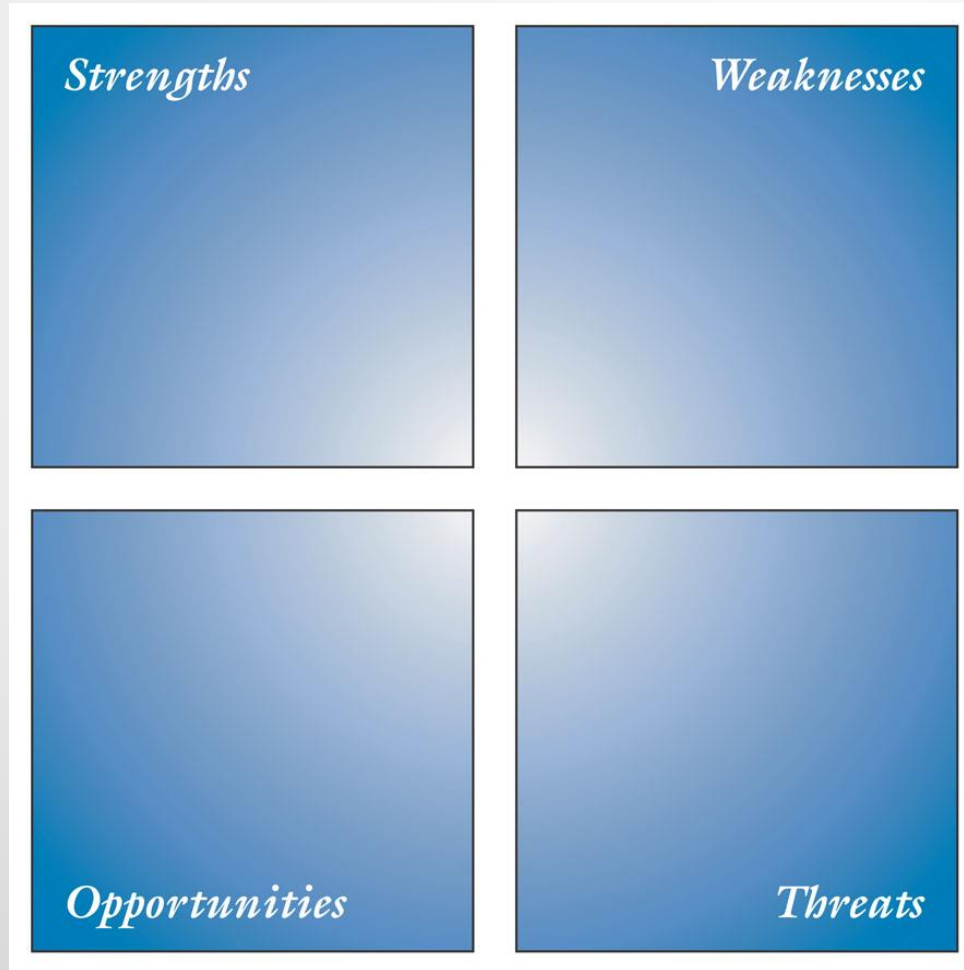


Risk Check List



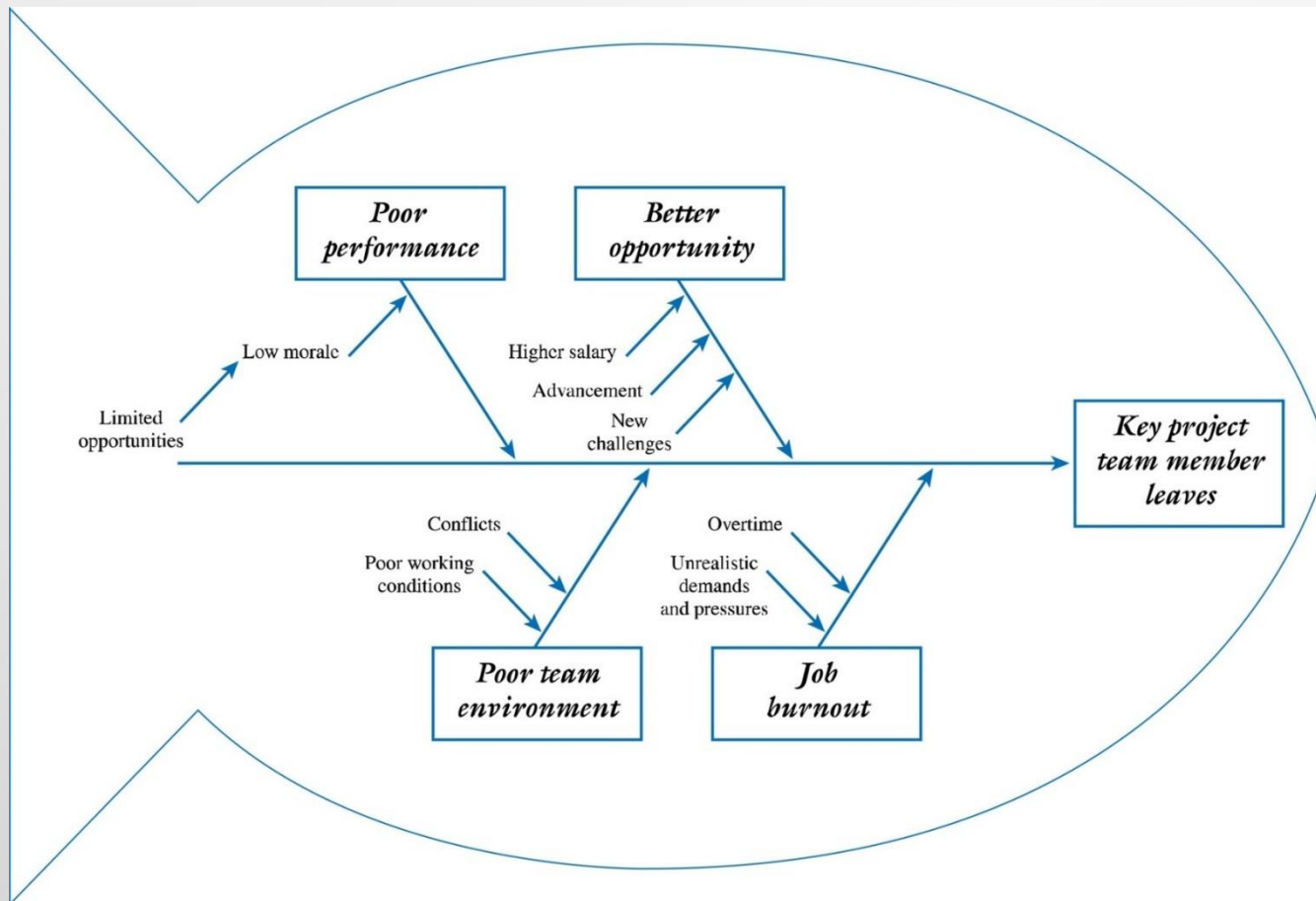
Fund	• Funding for the project has been secured • Funding for the project is sufficient • Funding for the project has been approved by senior management
Skills	• The project team has the requisite skills to complete the project
Manpower	• The project has adequate manpower to complete the project
Approvements	• The project charter and project plan have been approved • by senior management or the project sponsor
Realistic approach	• The project's goal is realistic and achievable • The project's schedule is realistic and achievable
Scope	• The project's scope has been clearly defined
Change	• Processes for scope changes have been clearly defined

SWOT Analysis



<https://www.youtube.com/watch?v=mRgeICQJLXA>

Cause & Effect Diagram



<https://www.youtube.com/watch?v=mLvizyDFLO4>

Risk Analysis & Assessment

$$\text{Risk} = f(\text{Probability} * \text{Impact})$$

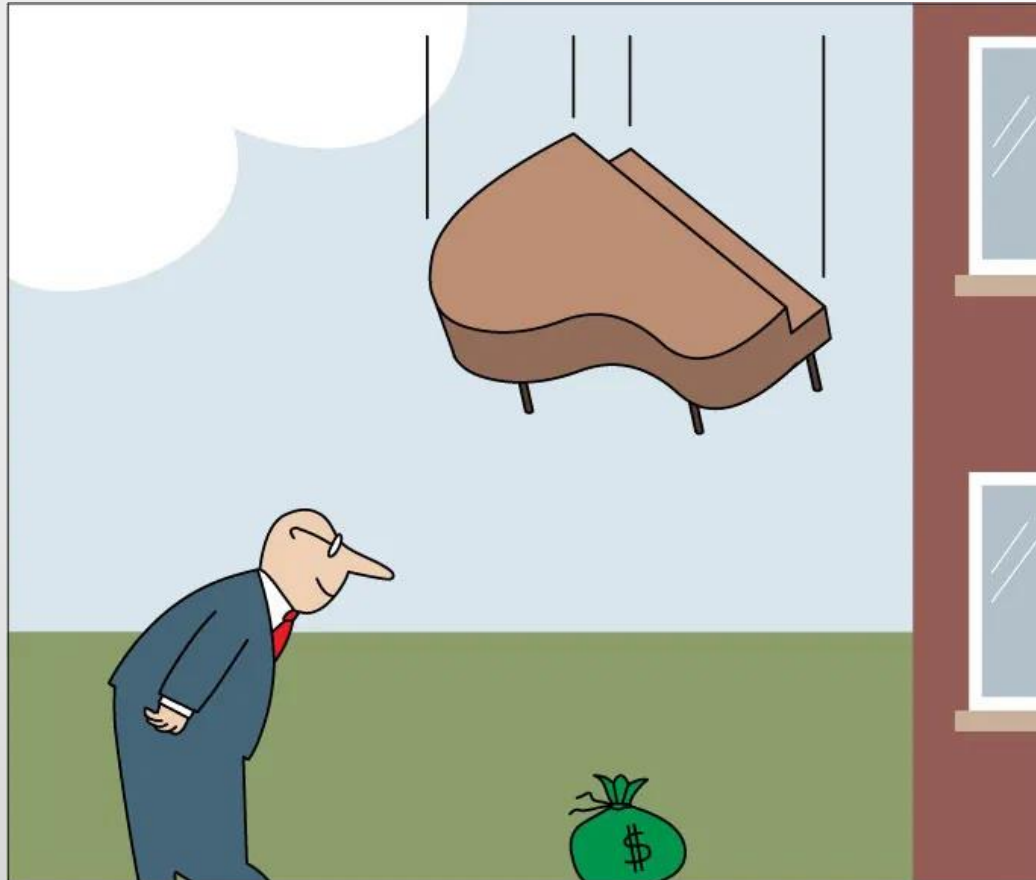
Depends on
Stakeholder risk
tolerances



Can't respond to all
risks!

Risk Analysis & Assessment

$$\text{Risk} = f(\text{Probability} * \text{Impact})$$



Tracking performance without risk is... shortsighted

Risk Analysis & Assessment

$$\text{Risk} = f(\text{Probability} * \text{Impact})$$


Risk assessment

- focuses on prioritizing risks

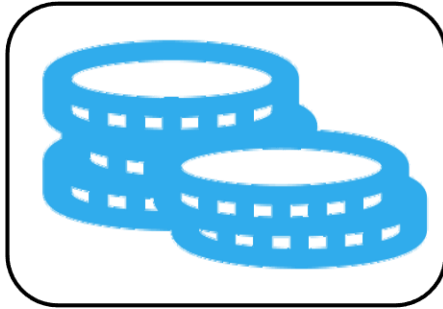


effective strategy

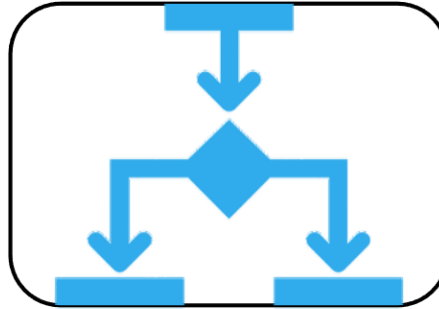
- to formulate for risks require a response.



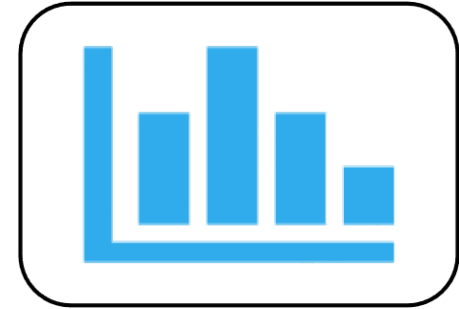
Qualitative Approaches:



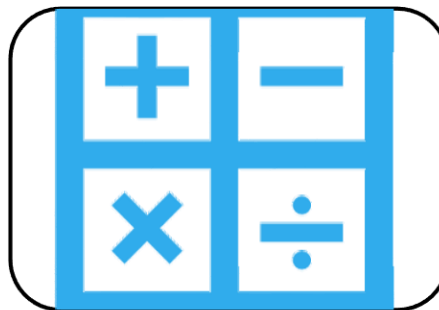
Expected Value
& Payoff Tables



Decision Trees



Risk Impact
Table & Ranking



Tusler's Risk
Classification

Payoff Table

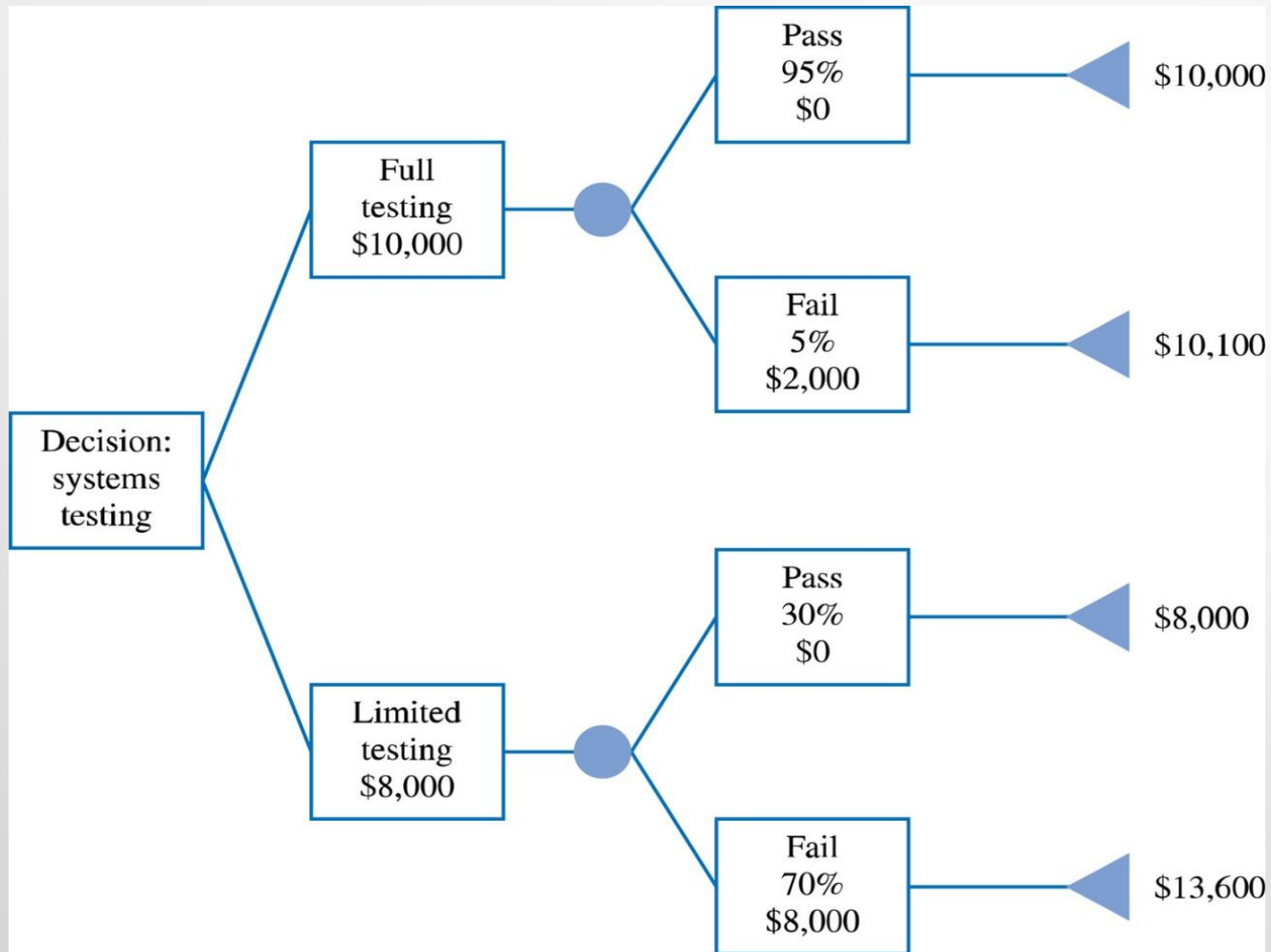
	A	B	A*B
Schedule Risk	Probability	Payoff (In thousands)	Prob * Payoff (In thousands)
Project completed 20 days early	5%	\$ 200	\$10
Project completed 10 days early	20%	\$ 150	\$30
Project completed on Schedule	50%	\$ 100	\$50
Project completed 10 days late	20%	\$ -	\$0
Project completed 20 days late	5%	\$ (50)	(\$3)
	100%		\$88

The
Expected
Value

Risks vs Return

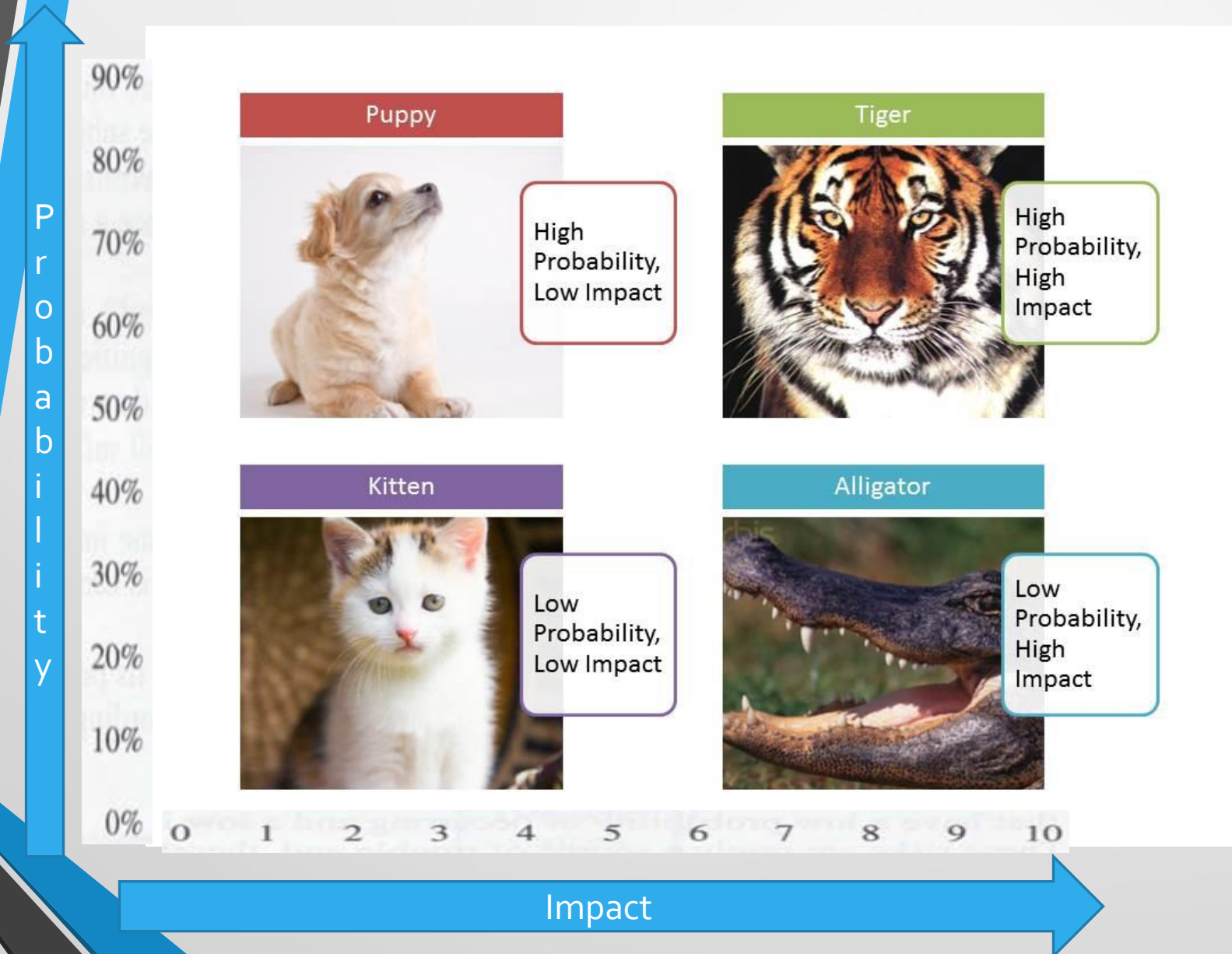
<https://www.youtube.com/watch?v=h7Fqk529BPo>

Decision Tree Analysis



Decision Analysis : EVSI - Expected Value of Sample Information
<https://www.youtube.com/watch?v=FUYo7dvaUuE>

Tusler's Risk Classification Scheme



Risk Impact



Risk Impact Table

Table 8.4 IT Project Risk Impact Analysis

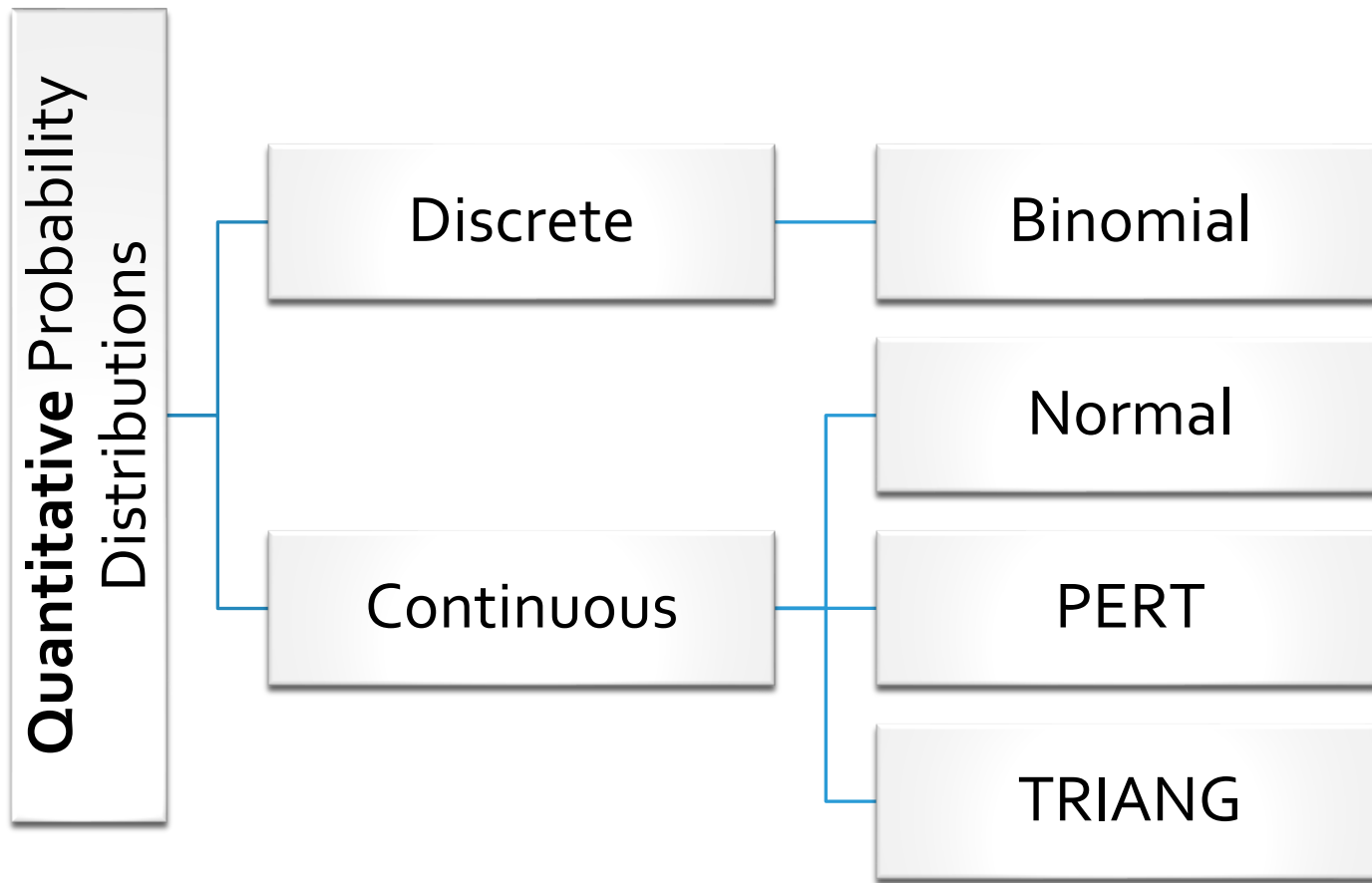
<i>Risk (Threats)</i>	<i>0–100% Probability</i>	<i>0–10 Impact</i>	<i>P • I Score</i>
Key project team member leaves project	40%	4	1.6
Client unable to define scope and requirements	50%	6	3.0
Client experiences financial problems	10%	9	0.9
Response time not acceptable to users/client	80%	6	4.8
Technology does not integrate with existing application	60%	7	4.2
Functional manager deflects resources away from project	20%	3	0.6
Client unable to obtain licensing agreements	5%	7	0.4

Risk Rankings

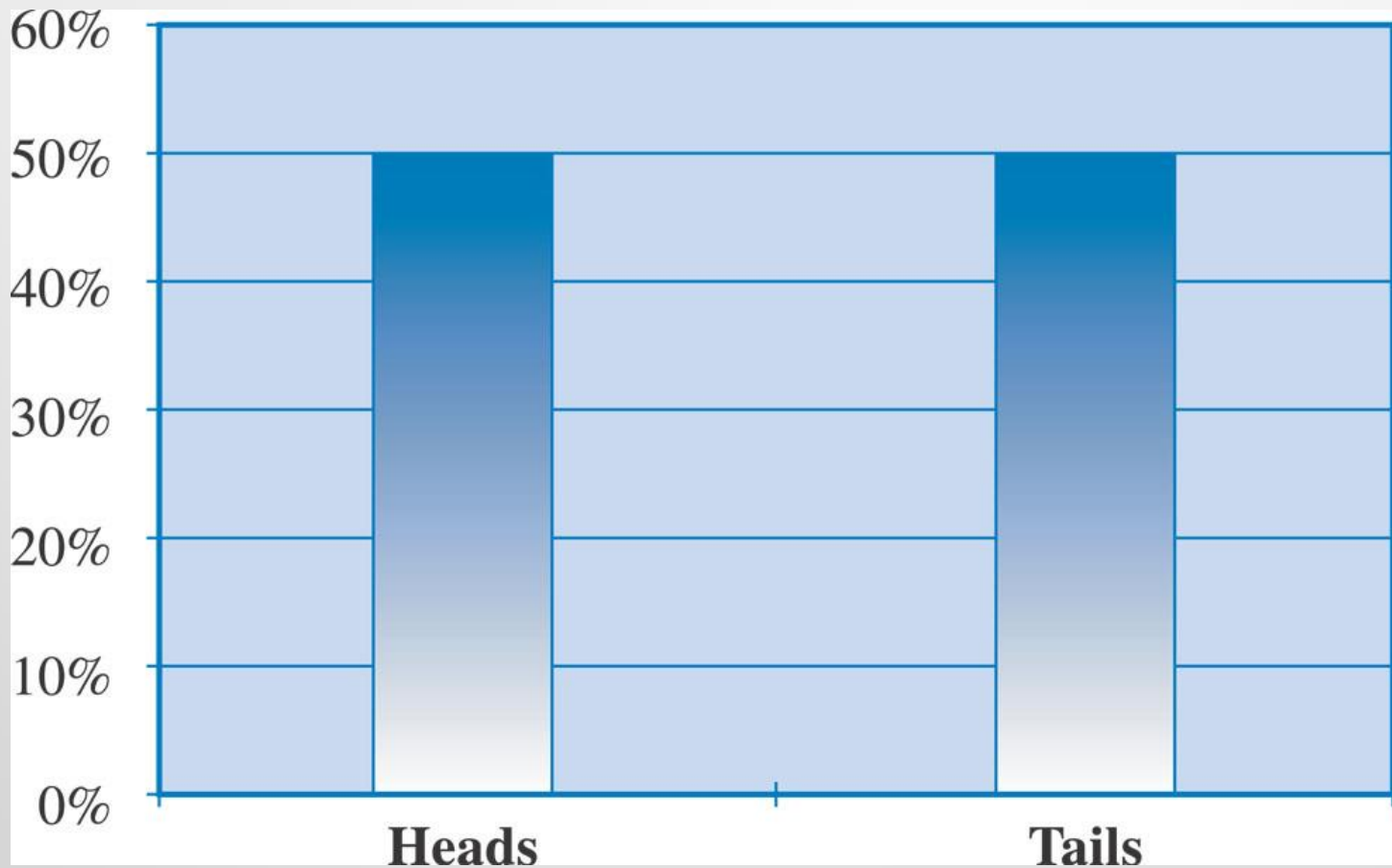
Table 8.5 Risk Rankings

<i>Risk (Threats)</i>	<i>Score</i>	<i>Ranking</i>
Response time not acceptable to users/client	4.8	1
Technology does not integrate with existing application	4.2	2
Client unable to define scope and requirements	3.0	3
Key project team member leaves project	1.6	4
Client experiences financial problems	0.9	5
Functional manager deflects resources away from project	0.6	6
Client unable to obtain licensing agreements	0.4	7

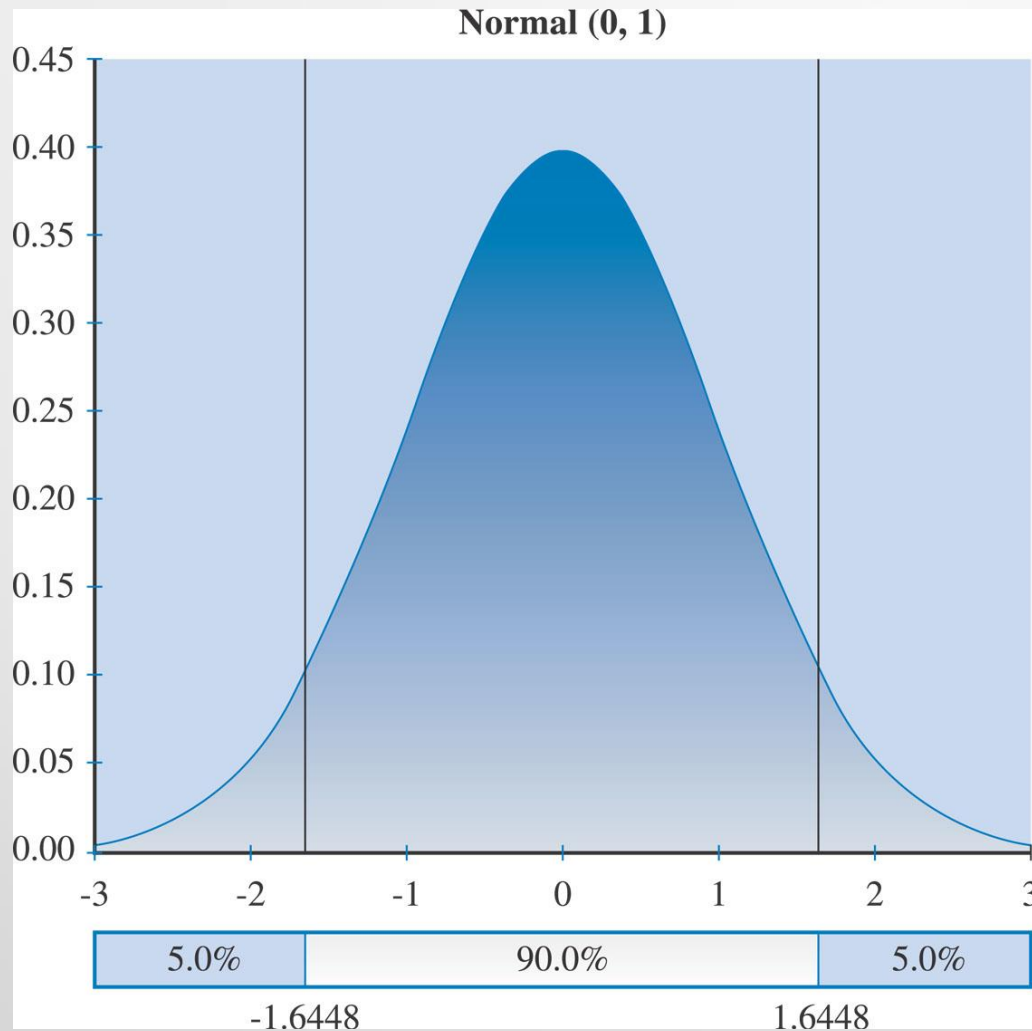
Risk Analysis & Assessment **Quantitative** Approaches



Binomial Probability Distribution



Normal Distribution

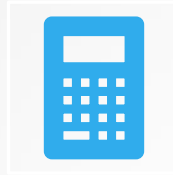


Normal Distribution



68% $\pm$

1 standard deviations of
mean



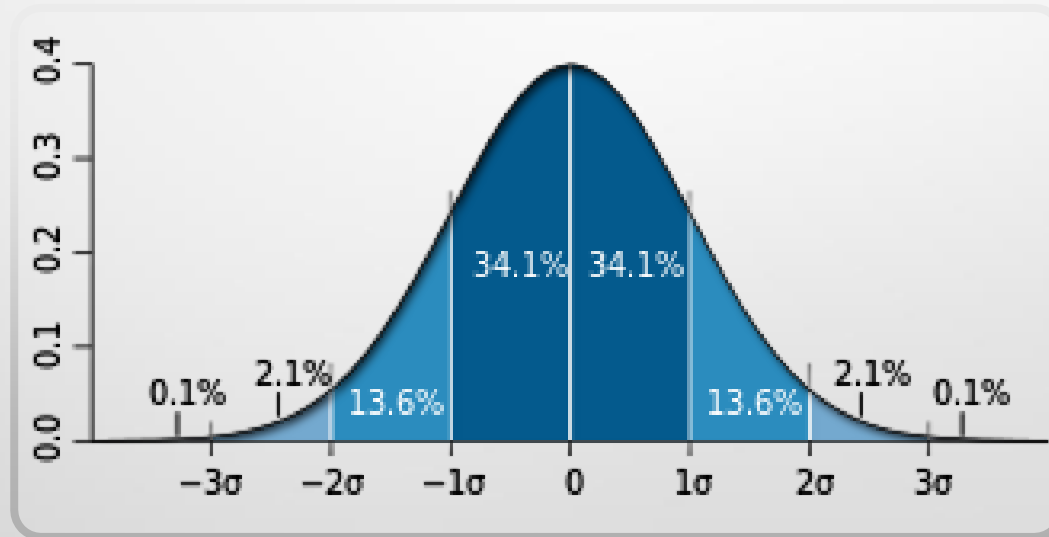
95%

± 2 standard deviations of
the mean



99%

± 3 standard deviations of
the mean



PERT DISTRIBUTION



PERT

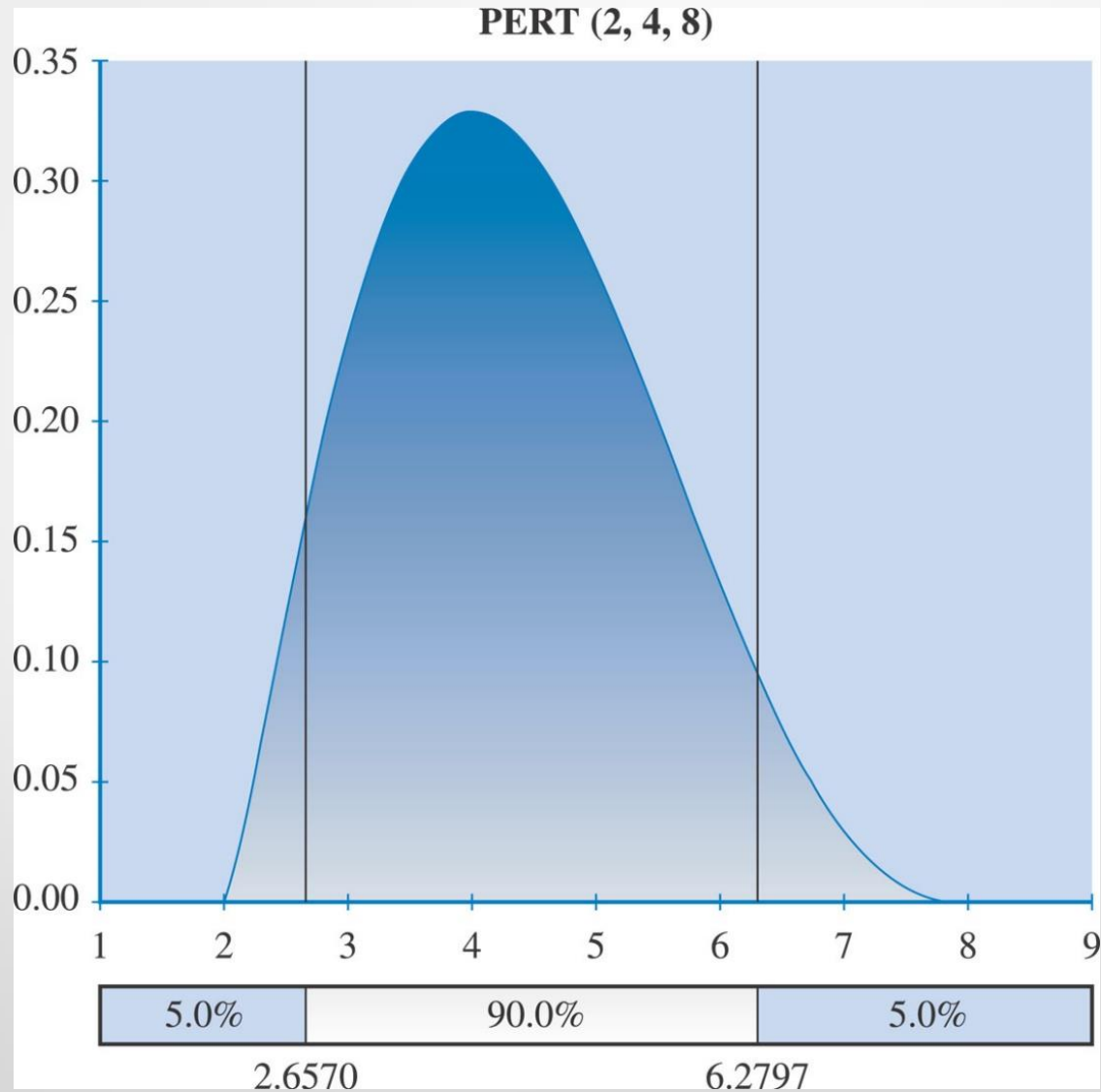
- $= (a + 4b + c)/6$



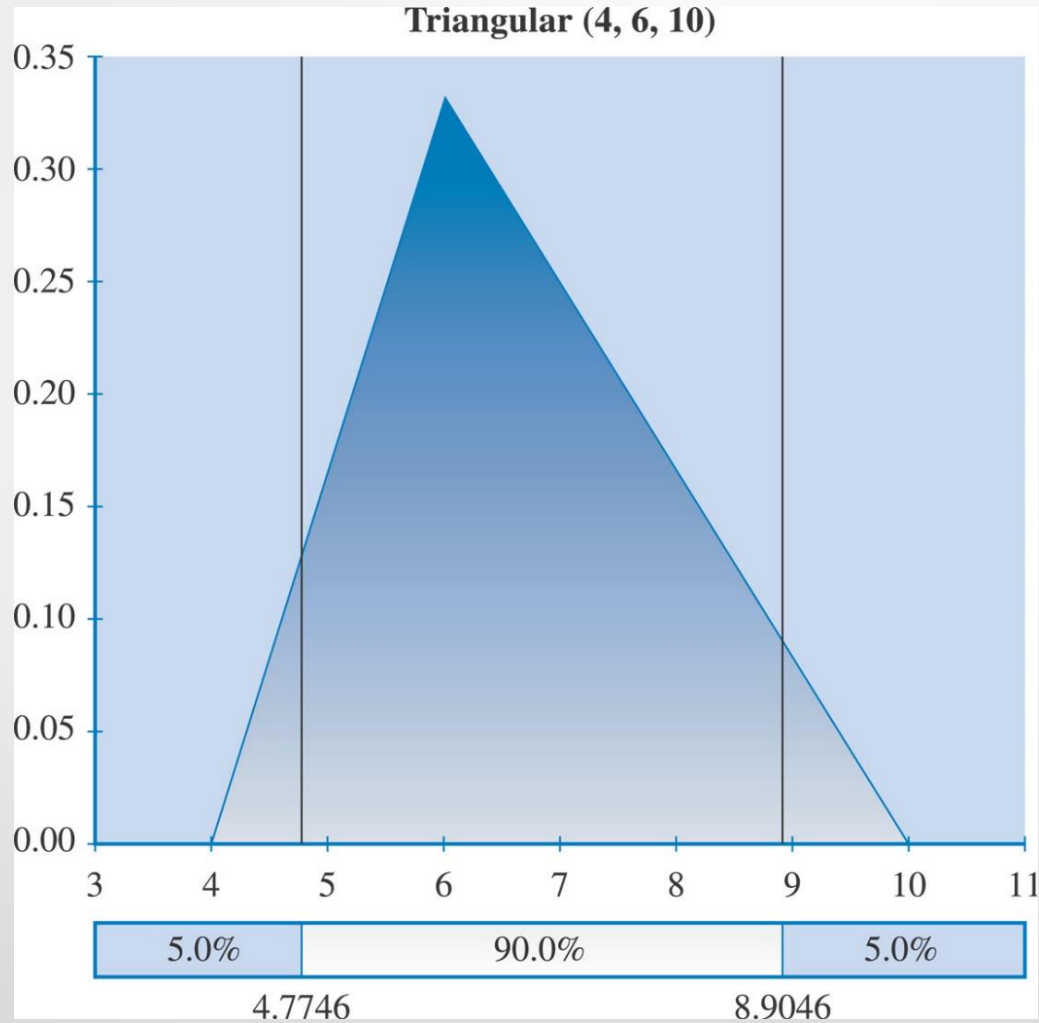
Legend

- a = optimistic estimate
- b = most likely
- c = pessimistic

PERT Distribution



Triangular Distribution



Simulations: Monte Carlo



Simulation Technique

- randomly generates specific values for a variable
- with a specific probability distribution



Iterations

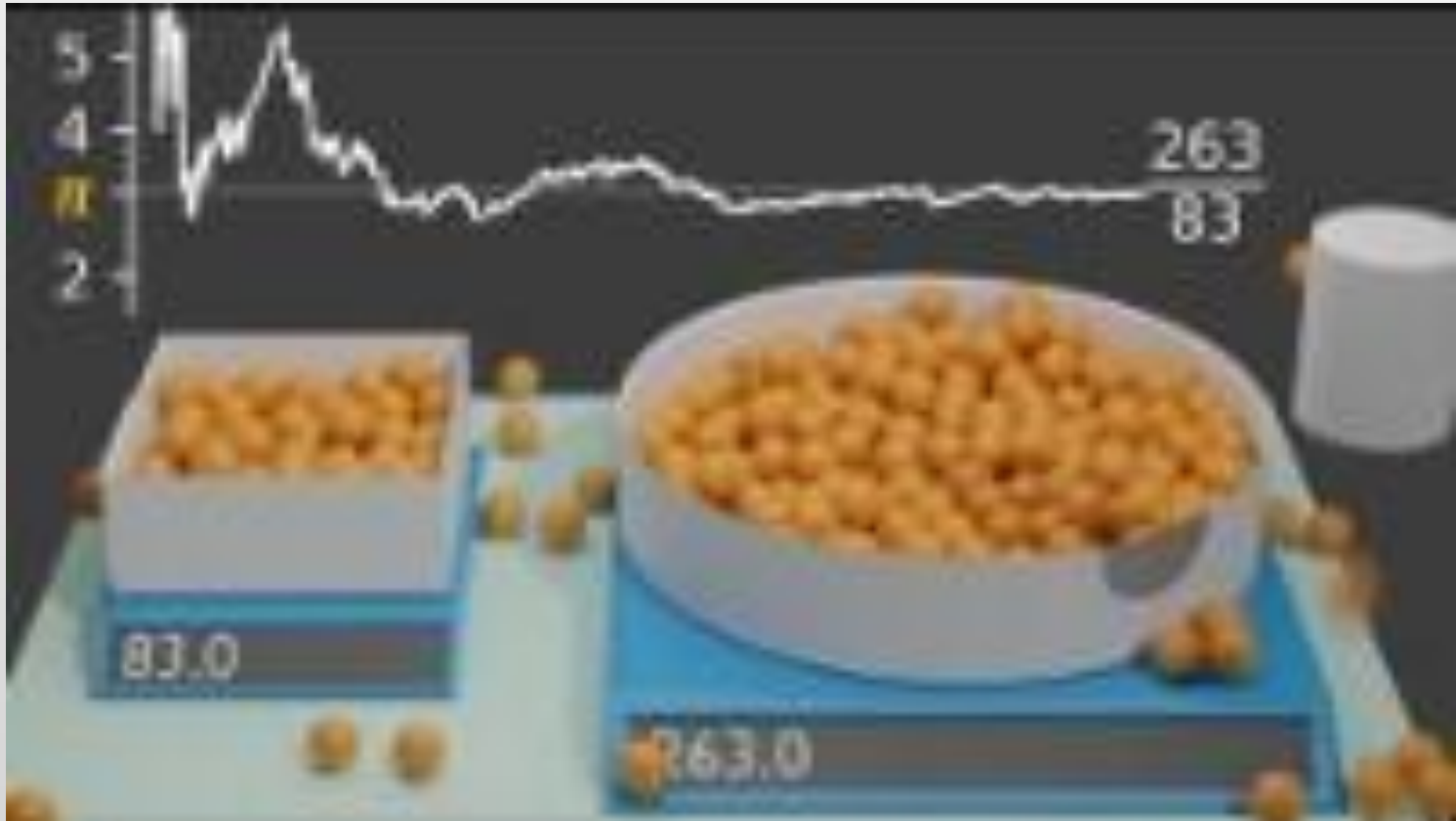
- Goes through a number of trials or iterations
- and records the outcomes



@RISK[®]

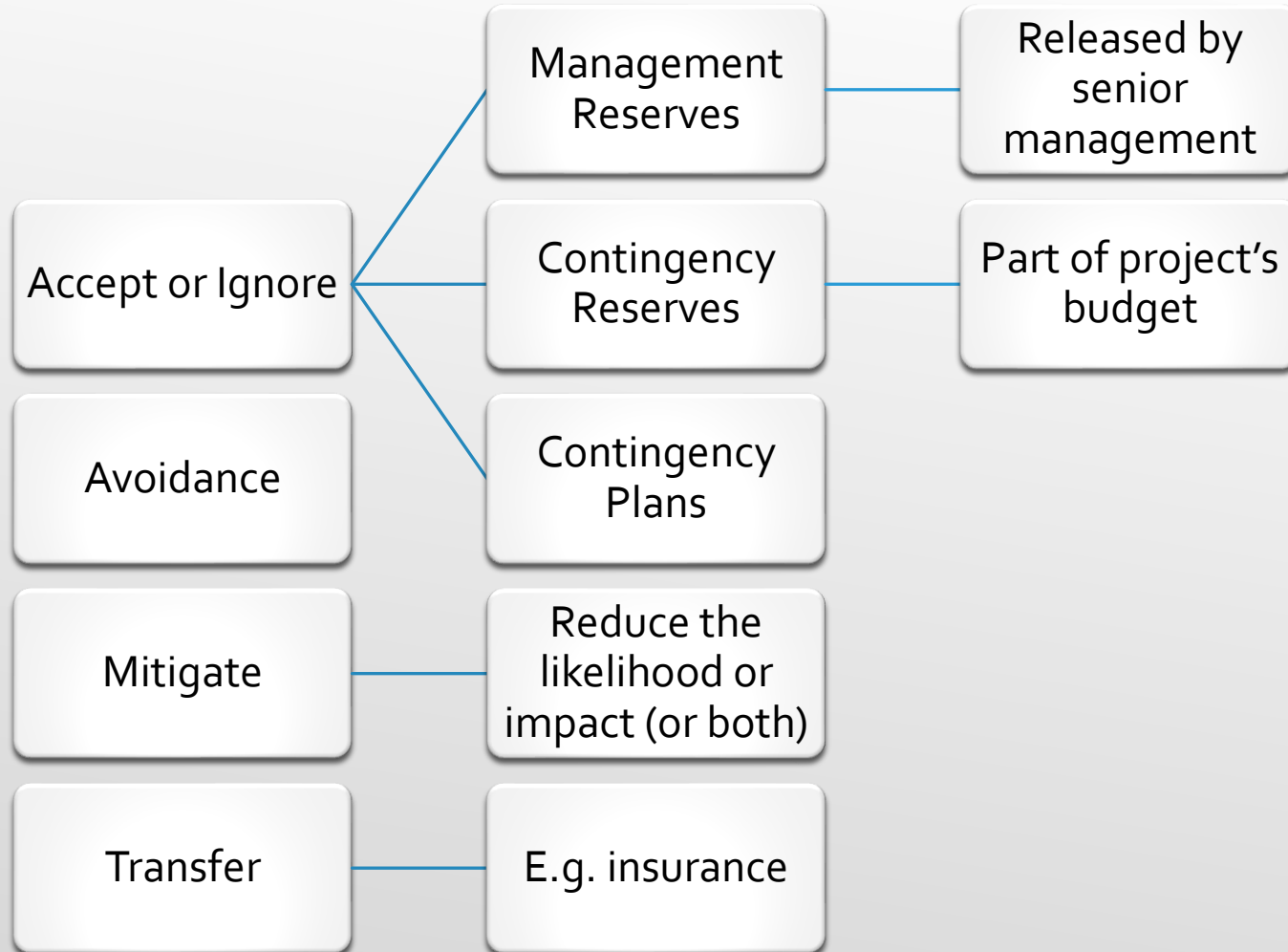
- An MS Project[®] add in that provides a useful tool
- for conducting risk analysis of your project plan

Monte Carlo Simulation



<https://www.youtube.com/watch?v=7ESK5SaP-bc>

Risk Strategies



Risk Strategies Depend On

The nature
of the risk

- Opportunity?
- Threat?

Impact

- on MOV and project objectives
- Probability?
- Impact?

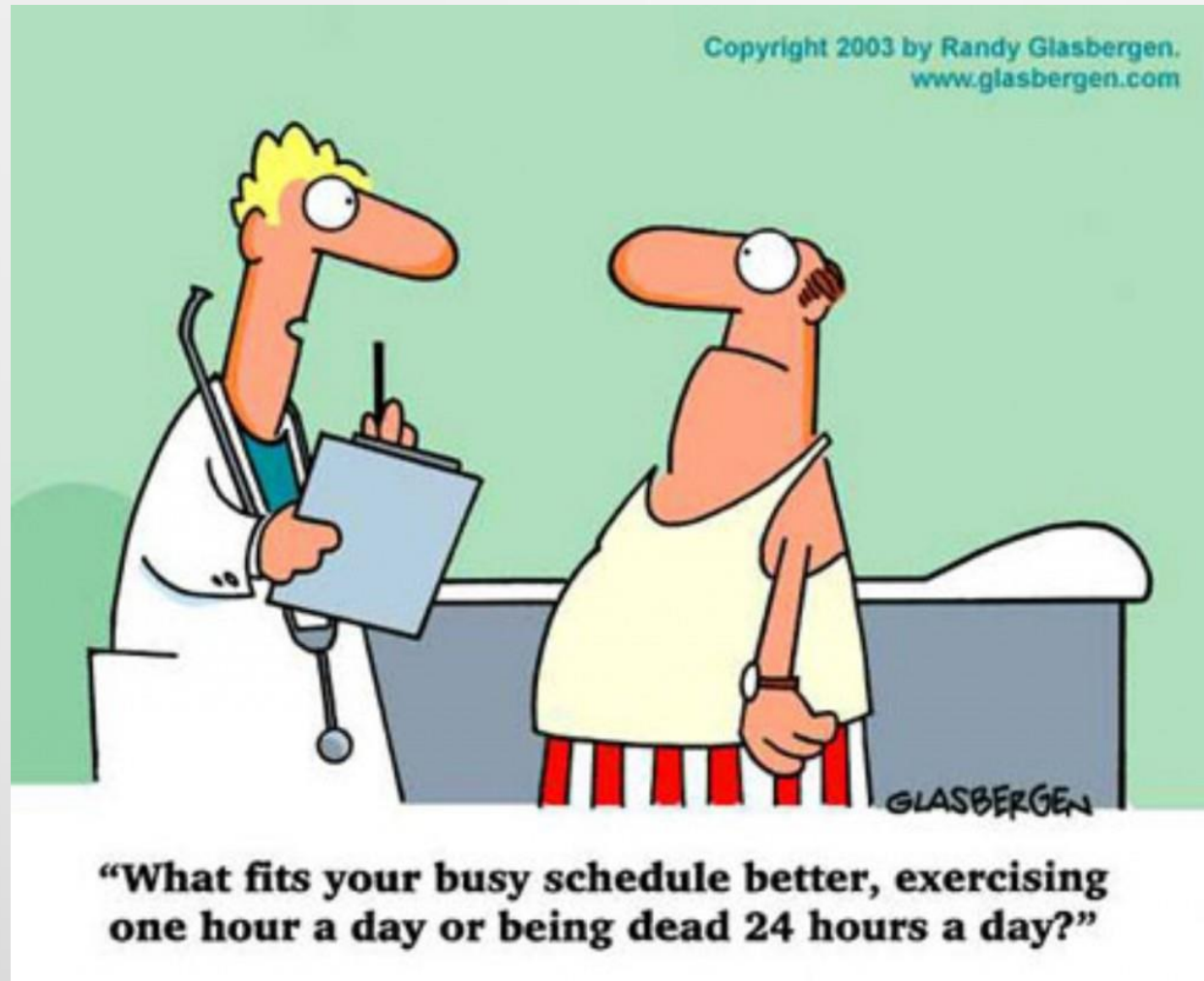
Project
constraints

- Resources
- Regulations
- Topologies
- Technologies
- Time-to-Market

...

Risk
tolerance

- Prevent
- Accept
- Ignore
- Mitigate



Risk Strategy Based on Trade-offs

Risk Monitoring & Control



Risk Audits

External to project team



Risk Reviews

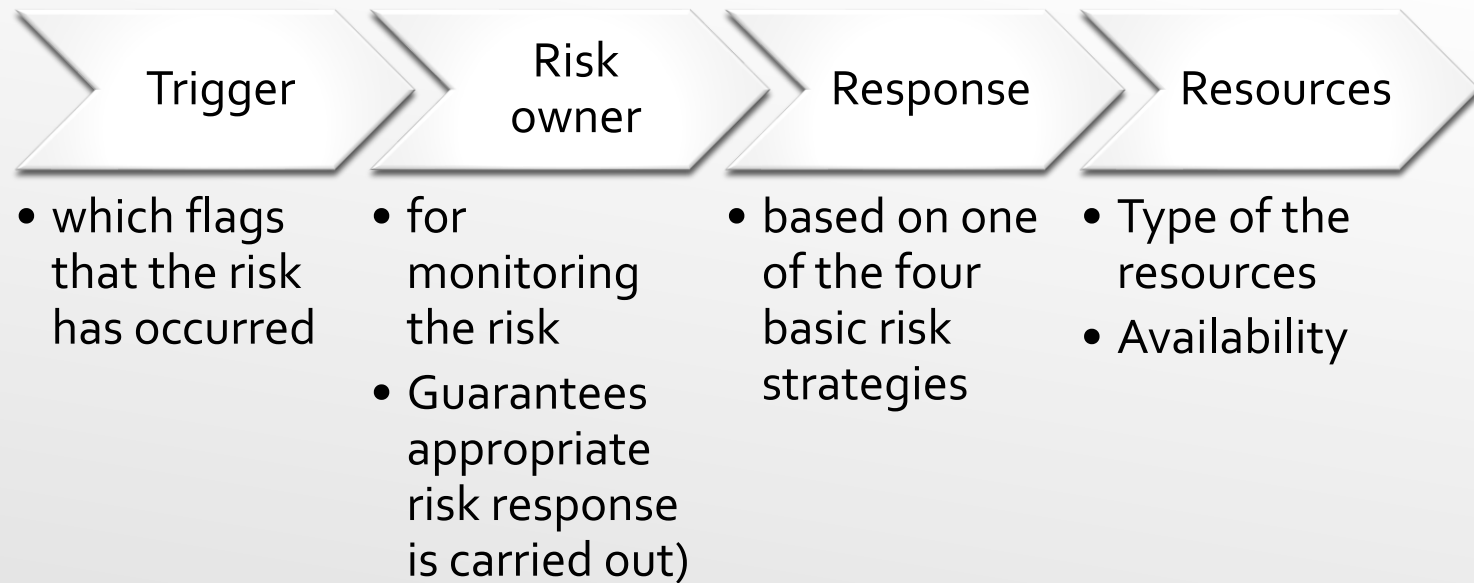
Internal



Risk Status

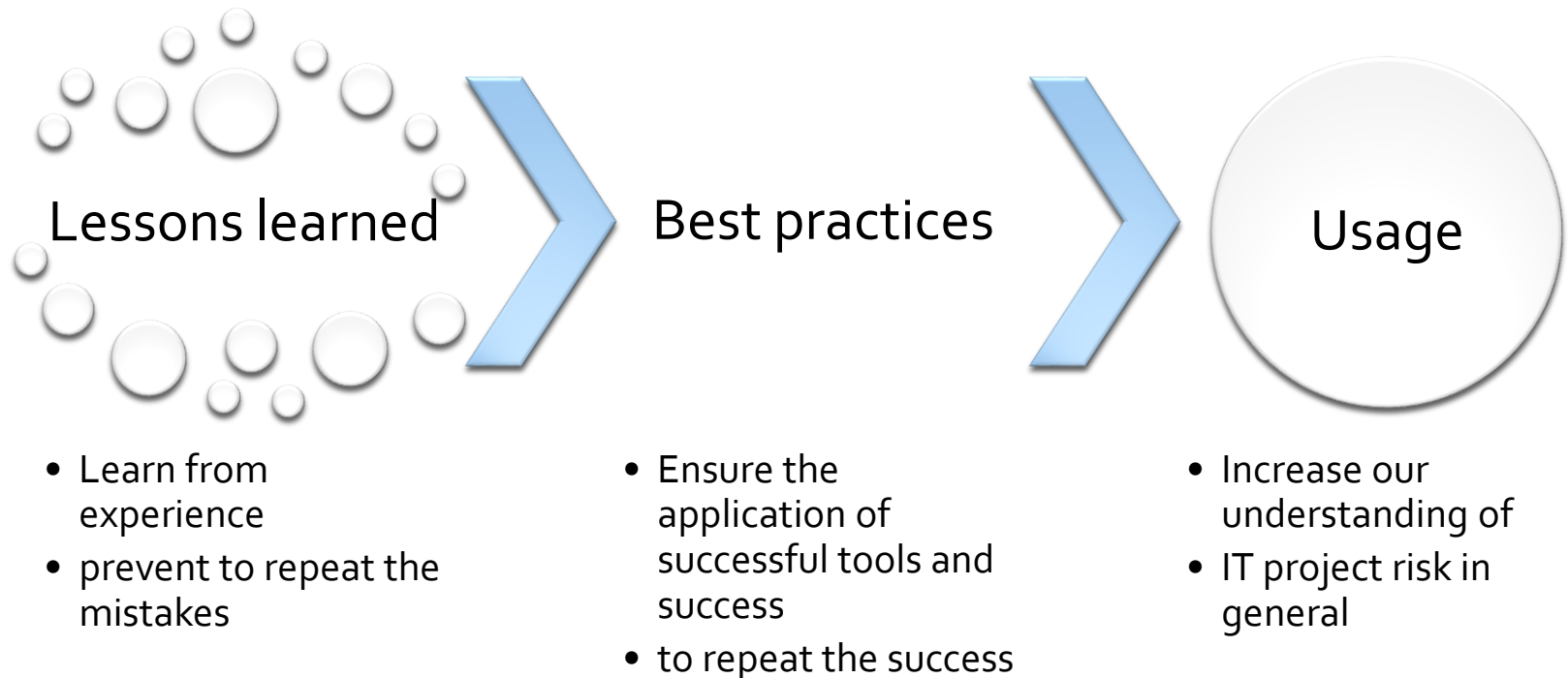
Risk Status Meetings
Reports

Risk Response Plan

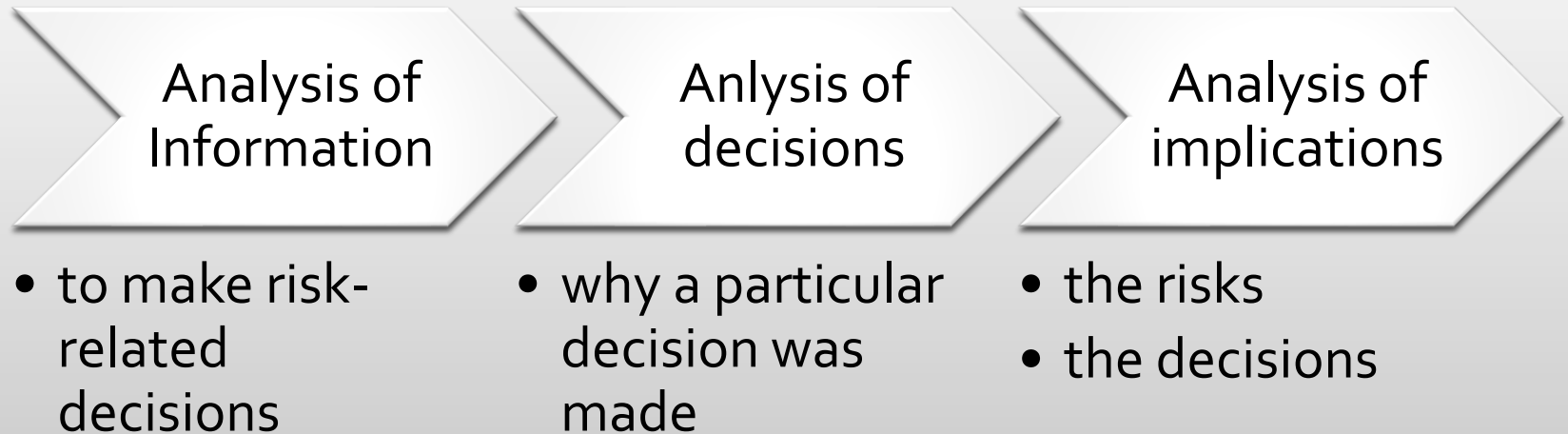


<i>Risk</i>	<i>Trigger</i>	<i>Owner</i>	<i>Response</i>	<i>Resources required</i>

Risk Evaluation



Risk Process Improvement



Self Assessment

- Asses what you got so far!

Noticable Issues

- What are the noticable issues in this chapter



Interesting Details

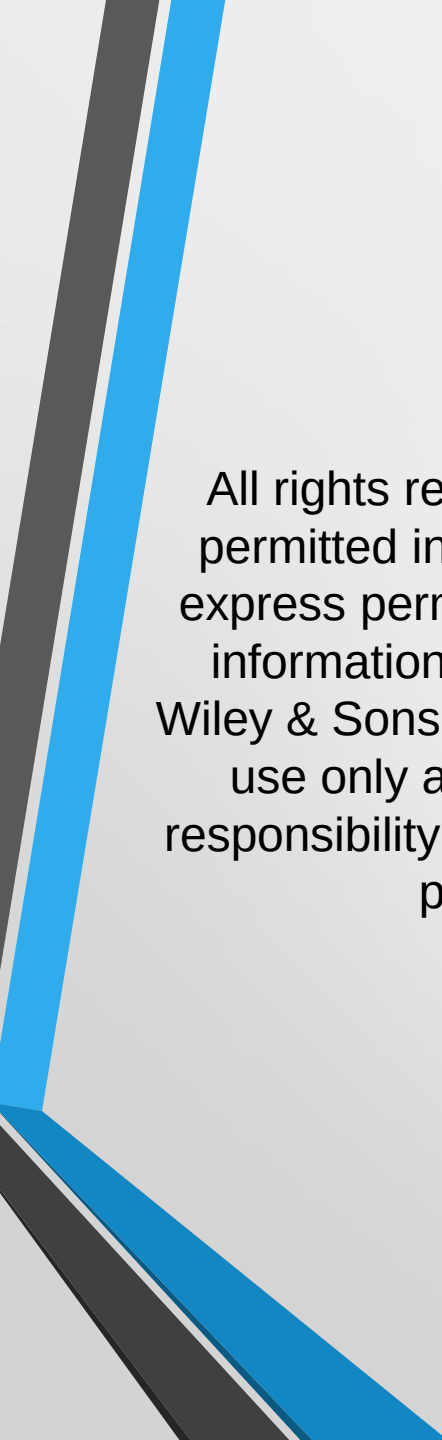
- What is the most interesting details in this chapter?



Unclarified issues

- What are the points need further explanations?





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